

ID	Title	Epic	User Story (Researcher POV)	Priority	Status	Acceptance Criteria	Functional Requirements	Non-Functional Requirements	Original Estimate (Days)
1	Normalized Research Data Warehouse	Database Design	As a Researcher, I want a structured research database to store API health data so that I can perform longitudinal analysis over several years.	Must	Ready for Dev	1. Schema supports Time-Series data.2. Relational mapping between disease types and regions.3. Successful indexing of historical data (>5 years).	• SQL-based data warehouse design. • Schema implementation for disease, climate, and demographics. • Partitioning strategy for time-based queries.	• Scalability: Support for 1TB+ data growth. • Reliability: 99.99% data persistence.	6
2	Automated ETL & Cleaning Pipeline	Data Preprocessing	As a Researcher, I want automated ETL pipelines so that API data is cleaned and transformed for analysis without manual intervention.	Must	In Progress	1. Automatic removal of duplicate API entries.2. Handling of missing values using mean/median imputation.3. Log of all cleaning steps.	• Python/Spark-based ETL scripts. • Outlier detection during ingestion. • Data type standardization (ISO dates, metric units).	• Performance: Processing of 100k records < 5 mins. • Reliability: Idempotent pipeline execution.	7
3	Data Validation & Normalization	Data Integrity	As a Researcher, I want validation and normalization mechanisms so that data from different global APIs is comparable and research accuracy is maintained.	Must	Backlog	1. Cross-API unit conversion (e.g., Fahrenheit to Celsius).2. Z-score normalization for ML features.3. Rejection of schemas not meeting health standards.	• Implementation of Min-Max scaling. • Unit conversion utility library. • Automated schema checking (Pydantic).	• Accuracy: 100% unit consistency across sources. • Performance: Real-time validation < 100ms.	5
4	Feature Engineering Module	ML Engineering	As a Researcher, I want feature engineering modules so that meaningful variables can be derived for ML models.	Should	Backlog	1. Module calculates rolling means and volatility.2. Lagged features created for time-series forecasting.3. Automated encoding of categorical region data.	• Feature store implementation. • Script for calculating incidence rates per 100k population. • One-hot encoding for demographic variables.	• Performance: Feature derivation < 1 sec per batch. • Scalability: Supports 500+ derived features.	8
5	Disease Trend Prediction Model	AI Analytics	As a Researcher, I want disease trend prediction models (LSTM/ARIMA) so that future outbreak patterns can be forecasted.	Must	Backlog	1. Model predicts next 30 days with < 15% MAPE error.2. Supports multi-disease forecasting.3. Visual comparison of predicted vs. actual trends.	• Implementation of LSTM (Deep Learning) and ARIMA. • Training pipeline using historical API data. • Forecast API endpoint for frontend consumption.	• Performance: Model inference < 2 seconds. • Reliability: Model retraining every 24 hours.	8
6	AI Anomaly Detection Algorithm	AI Analytics	As a Researcher, I want anomaly detection algorithms so that unusual spikes in public health data are identified.	Must	Backlog	1. Identification of spikes outside 3 standard deviations.2. Real-time alerting for detected anomalies.3. Low false-positive rate (< 5%).	• Isolation Forest or Autoencoder implementation. • Threshold-based alerting logic. • Metadata tagging for anomalous data points.	• Performance: Detection within < 500ms of ingestion. • Reliability: 99% recall for known outbreak spikes.	7
7	Model Evaluation & Validation	ML Engineering	As a Researcher, I want model performance evaluation metrics so that I can validate prediction reliability before publishing findings.	Should	Backlog	1. Automated calculation of RMSE, MAE, and R-squared.2. Generation of Confusion Matrices for classification tasks.3. Log of hyperparameter tuning results.	• Metric logging via MLFlow or similar tool. • Automated Cross-validation scripts. • Accuracy report dashboard for researchers.	• Accuracy: Evaluation metrics calculated to 4 decimal places. • Reliability: Traceable model versioning.	5
8	Automated Research Report Gen	Research Tools	As a Researcher, I want automated research report generation so that findings can be documented efficiently.	Should	Backlog	1. PDF export containing ML metrics and trend charts.2. Summary of data sources and processing steps.3. Automated bibliography of API sources used.	• LaTeX or PDFKit integration for report rendering. • Dynamic data insertion from the ML module. • Template-based report structuring.	• Usability: Report generation < 10 seconds. • Portability: PDF/A compliant documents.	4
9	Scalable Big Data Architecture	Database Design	As a Researcher, I want a scalable database architecture so that increasing API data volumes can be handled.	Should	Backlog	1. Database horizontal scaling (sharding) tested.2. Query performance remains stable at 10M+ records.3. Automated backup and recovery verified.	• Implementation of database clustering. • Read/Write splitting (Load balancing). • Automated database index maintenance.	• Scalability: Support for 100M+ records. • Performance: Complex query execution < 3 sec.	7
10	Secure Data Integrity & Compliance	Security	As a Researcher, I want secure research data storage so that data integrity and research compliance are ensured.	Must	Backlog	1. Database level encryption (TDE).2. Audit trails for every database modification.3. Role-based access for DB views.	• Transparent Data Encryption (TDE). • Logging of SQL "Select/Update" queries. • Multi-factor authentication for DB access.	• Security: SOC2/HIPAA compliance. • Reliability: 100% data auditability.	5
11	Statistical Analysis Module	Statistical Analysis	As a Researcher, I want statistical analysis modules so that hypothesis testing can be performed.	Must	Ready for Dev	1. Automated p-value calculation for correlations.2. Correlation matrix for all variables.3. T-test/ANOVA support for comparative studies.	• Integration with SciPy/Statsmodels libraries. • Calculation of Pearson and Spearman correlations. • Automated significance testing.	• Accuracy: High precision floating-point math. • Performance: Full correlation matrix < 2 sec.	6
12	Distributed Query Optimization	Performance	As a Researcher, I want optimized query performance so that my large-scale analysis scripts don't timeout.	Could	Backlog	1. Query execution time reduced by 50% through caching.2. Implementation of Materialized Views.3. Automated slow-query detection.	• Redis-based query caching layer. • Materialized view implementation for aggregates. • Database query plan optimization.	• Performance: 90% of queries < 1 second. • Reliability: No timeouts for 15-min scripts.	6